

Patient Shortage in Histopathology: Breadth, Depth, and Effective Support

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Abstract

Training on fewer patients impairs histopathology patch classification even when class patch counts are held fixed, and neither a different downstream classifier nor an equalized training objective recovers the loss. This experiment separates patient breadth from patch depth on a three-by-three allocation grid in BRACS and TCGA-UT, using frozen Virchow2 features and fixed patient-disjoint splits. All cells draw from common maximum-depth patient pools; fixed-breadth depth comparisons use the same patients and nested patches. At 160 patches per class, allocating eight patches to each of twenty patients instead of 32 patches to each of five patients improves patient-macro balanced accuracy by 3.71 percentage points on BRACS (95% interval: 2.21 to 5.35) and 9.52 points on TCGA-UT (8.96 to 10.07). Redundancy-adjusted effective support describes the BRACS grid best among the single-predictor models, but its residual breadth interval does not establish practical equivalence to zero. On TCGA-UT, additional depth produces gains below one point, while a positive residual breadth coefficient remains after adjustment for effective support. The findings favour broader patient sampling at the tested budget and identify a limitation of scalar redundancy adjustment on TCGA-UT; they do not identify the biological or acquisition variation responsible.

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1 Motivation

The signal taxonomy of Queisler (2026c) separates *nominal support*, the number of labelled patches n_c available for class c , from *independent support*, the number of distinct patients G_c supplying them. Patches from one patient share acquisition conditions and much of their morphology, so a large patch count assembled from few patients need not carry the evidence its size suggests. The same taxonomy already anticipates a quantitative version of this argument. Its *redundancy-adjusted effective support* $N_{\text{eff},c} = n_c / \text{DE}_c$ divides the patch count by a design effect $\text{DE}_c = 1 + (m_c^* - 1) \text{ICC}_c$, where m_c^* is the size-biased number of patches per patient and ICC_c is the within-patient correlation of a scalar summary of the representation (Kish, 1965; Rutterford et al., 2015).

Three earlier results define what is left to explain. The controlled shortage benchmark reduced patient support while holding class patch counts fixed and measured discrimination losses of 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT, which the tested mitigation methods recovered by between -6.5% and 2.5% (Queisler, 2026b). The classifier comparison then showed that the deficit is not a property of the downstream model: replacing the multilayer perceptron by multinomial logistic regression or cosine nearest neighbours improved concentrated-support accuracy by at most 0.48 points, while broader coverage benefited every classifier (Queisler, 2026e). The influence comparison finally showed that the deficit is not an artefact of unequal loss contributions: patient-average training changed accuracy by -1.36 points on BRACS and -0.004 points on TCGA-UT, leaving residual coverage benefits of 2.73 and 4.70 points (Queisler, 2026d). The bottleneck has thus been localized to the training material itself, but its content has not been characterized.

Two readings of that material remain. Under *within-patient redundancy*, patches from one patient are statistically dependent, so a patient’s contribution is worth fewer independent observations than its patch count states; the deficit is then a shortage of effective observations and should be summarized by a single scalar. Under *coverage beyond redundancy*, patients differ in ways that no scalar correlation captures, because additional patients introduce morphological or acquisition modes absent from the observed ones; the deficit is then a shortage of distinct conditions rather than of effective observations. These readings are not mutually exclusive, and both are compatible with every result reported so far, because the previous allocations varied patient count and within-patient depth together.

Separating them requires an allocation in which the number of contributing patients and the number of patches taken from each of them move independently. This experiment therefore constructs a *breadth* $\times$ *depth* grid, with breadth G the number of patients contributing to a class and depth m the number of patches drawn from each of them, and compares allocations that differ in composition at an identical patch budget. Holding depth equal across patients has a second benefit: it removes the contribution inequality studied by Queisler (2026d) by construction, so patch-average and patient-average training coincide and the two objectives need not be compared again.

2 Study design

BRACS retains its seven lesion classes and TCGA-UT its 30 eligible cancer types. Both keep the benchmark dataset eligibility rules, class order, and frozen 2560-dimensional Virchow2 features of the benchmark, formed by concatenating the CLS token and the mean patch token (Zimmermann et al., 2024; Queisler, 2026a). The three existing patient-disjoint train–validation–test splits remain fixed, and BRACS patients and TCGA-UT participants are called patients throughout. Validation and test partitions keep their natural distributions and are never resampled.

Each training allocation is described by one pair (G, m) applied to every class: exactly G patients contribute to class c , and each contributes exactly m patches of that class, so $n_c = Gm$

for all classes. Class patch counts are therefore equal by construction, which removes prevalence as a competing signal and leaves support as the only quantity that varies. Table 1 lists the nine cells of the grid and their per-class patch budgets. Breadth ranges over $G \in \{5, 10, 20\}$ and depth over $m \in \{8, 16, 32\}$, a fourfold range in each factor and a sixteenfold range in n_c . The largest cell uses 4,480 training patches on BRACS and 19,200 on TCGA-UT.

Patients per class G	Patches per patient m		
	8	16	32
5	40	80	160
10	80	160	320
20	160	320	640

Table 1: Per-class patch budget $n_c = Gm$ for the nine allocation cells. Cells sharing a budget form equal-budget sets: $n_c = 160$ contains (20, 8), (10, 16), and (5, 32), while $n_c = 80$ and $n_c = 320$ contain two cells each. Comparisons within such a set differ only in how a fixed number of patches is distributed over patients.

The grid ceiling is set by the rarest class rather than by cost. The benchmark’s broad-coverage allocations contain at least 30 patients per class, but this does not establish how many can supply 32 patches in each training split (Queisler, 2026d). A separate audit confirmed feasibility of the common maximum-depth pools (Section 5). The largest budget remains below the benchmark’s balanced allocations of about 2,400 patches per class on BRACS and about 1,494 on TCGA-UT. Absolute accuracies are not directly comparable to those earlier reports; the object of interest is the shape of the support surface.

2.1 Allocation rules

Let N_{irc} denote the number of available patches of class c from patient i in the training partition of split r . Eligibility is fixed before any cell is sampled:

$$\mathcal{E}_{rc} = \{i : N_{irc} \geq m_{\max}\}, \quad m_{\max} = \max\{8, 16, 32\} = 32. \quad (1)$$

Every breadth–depth cell uses this same class–split pool, including depths 8 and 16. Patients with fewer than 32 patches of the class are excluded from all cells. Otherwise, changing depth could change the distribution of tissue availability, lesion extent, slide count, or morphology among eligible patients, confounding a depth contrast with patient composition.

Before fitting, the preflight audit records $|\mathcal{E}_{rc}|$ for every class and split and requires $|\mathcal{E}_{rc}| \geq 20$ throughout. If this fails, execution stops and the entire three-by-three grid must be reduced consistently before a fresh preflight; no class, split, or cell receives a separate eligibility threshold or allocation ladder. Breadth is reduced first while retaining the common 32-patch threshold. If the maximum depth must also be reduced, eligibility is redefined once at that maximum for every cell. Realized ladders, pool counts, patch budgets, and contrast endpoints must be documented before fitting.

For each split, class, breadth G , and repeated draw d , G patients are sampled without replacement from $\mathcal{E}_{rc}$. The patient seed does not depend on the cell depth, so these exact patients contribute at all three depths within the draw. Classes are sampled independently; a patient eligible for several classes may contribute to more than one. For each selected patient, patches follow one deterministic round-robin ordering across slides, with stable slide and patch ordering. Each depth reveals a prefix of that same ordering, giving

$$\mathcal{P}_{ircd,8} \subset \mathcal{P}_{ircd,16} \subset \mathcal{P}_{ircd,32}. \quad (2)$$

Thus a fixed-breadth depth contrast changes only how many patches are revealed from the same patients under this selection rule. Each cell uses $R = 5$ repeated patient draws; every cell quantity averages over these draws, with corresponding draw indices paired across depths. Between-draw dispersion is reported separately as a descriptive measure of sensitivity to patient composition.

The readout is the deterministic multinomial logistic regression of Queisler (2026e), with an unpenalized intercept and the same feature preprocessing and convergence criteria. Its regularization strength is selected on the validation partition separately for every split, cell, and draw from the inherited grid, because the training-set size varies by a factor of sixteen across the grid and a single inherited value would confound support with regularization. Only converged fits enter the analysis. The multilayer perceptron and the nearest-neighbour readout are omitted: Queisler (2026e) established that the coverage effect is present under all three, and a single deterministic readout keeps the grid small enough to be run without repeated initializations.

3 Effective support

Equal depth simplifies the redundancy adjustment considerably. With every contributing patient supplying m patches, the size-biased patient size $m_c^* = \sum_g m_{gc}^2 / n_c$ equals m exactly, so the design effect and the effective support of Queisler (2026c) become

$$\text{DE}_c(m) = 1 + (m-1) \text{ICC}_c, \quad N_{\text{eff},c}(G, m) = \frac{Gm}{1 + (m-1) \text{ICC}_c}, \quad \lim_{m \rightarrow \infty} N_{\text{eff},c} = \frac{G}{\text{ICC}_c}. \quad (3)$$

The two grid factors therefore enter effective support asymmetrically. Breadth scales it proportionally, whereas depth scales it only until the design effect grows with m , after which additional patches from the same patients contribute progressively less. The limit makes the prediction concrete: under the redundancy account no depth can raise effective support beyond G/ICC_c , and the transition occurs near $m \approx 1/\text{ICC}_c$. Whether that transition is visible within the depth ladder is an empirical question about the representation, and one that the estimated correlations settle before any classifier is fitted.

3.1 Estimating the intraclass correlation

The correlation is estimated per class and split from the training partition only, following the procedure fixed in the taxonomy. Each frozen Virchow2 embedding is projected onto the leading principal direction of a principal component analysis fitted jointly across classes on a class-balanced reference sample, which yields a scalar score that does not depend on any fitted classifier. The one-way random-effects estimator with the unequal-cluster-size correction is then applied to these scores with patients as clusters (Kenny and La Voie, 1985), from a capped sample of at most 200 patients and 200 patches per patient. Class correlations are averaged over the three training splits before evaluating Equation (3); the cell predictor then averages the resulting $N_{\text{eff},c}$ over classes. Correlations characterize the training partitions rather than only the maximum-depth eligible pools.

Reporting these correlations serves a purpose beyond the support surface. Each ICC_c is the share of projected feature variance that lies between patients rather than within them, so the class-level table is a direct measurement of intra- and inter-patient heterogeneity for every lesion type and cancer type. Classes whose appearance varies strongly within a patient should have low correlations and lose comparatively little from concentrated sampling, and comparing these estimates with the class-level accuracy effects is an exploratory part of the analysis.

Equation (3) lets the three accounts of Section 1 be stated as separable predictions about the grid, summarized in Table 2. The predictions are distinguishable because $n = Gm$, G , and N_{eff} are different functions of the two grid factors: an equal-budget set holds n constant while G and N_{eff} still vary across its cells.

Account	Claim about the deficit	Prediction on the grid
Nominal support	Only the amount of labelled material matters	Accuracy is a function of $n = Gm$; equal-budget cells agree
Redundancy-adjusted support	Within-patient patches are correlated, so they are worth fewer independent observations	Accuracy is a function of N_{eff} ; depth saturates near $m \approx 1/\text{ICC}$; cells collapse onto one curve
Coverage beyond redundancy	Patients supply variation that a scalar correlation cannot express	Breadth remains beneficial at matched N_{eff} ; a positive residual breadth term persists

Table 2: The three accounts and the grid behaviour each of them implies. The accounts are ordered by how much structure they attribute to the patient grouping. The first is already disfavoured by the benchmark result and is retained as a reference line within this design.

4 Evaluation

The primary endpoint is *patient-macro balanced accuracy*, defined exactly as in the two preceding experiments so that effect sizes remain comparable. For split r , let $\mathcal{P}_{rc}$ contain the test patients with at least one patch of true class c , and let $\mathcal{J}_{irc}^{\text{test}}$ contain that patient’s evaluated patches of that class. The split score and the underlying within-patient recalls are

$$A_r = \frac{1}{K} \sum_{c=1}^K \frac{1}{|\mathcal{P}_{rc}|} \sum_{i \in \mathcal{P}_{rc}} q_{irc}, \quad q_{irc} = \frac{1}{|\mathcal{J}_{irc}^{\text{test}}|} \sum_{j \in \mathcal{J}_{irc}^{\text{test}}} \mathbf{1}\{\hat{y}_j = c\}, \quad (4)$$

where K is the number of classes. A cell score $A(G, m)$ averages the three split scores and the R draws equally. Secondary endpoints are macro negative log-likelihood in nats and expected calibration error over ten equal-width confidence bins, which connect this grid to the probability-quality damage established by Qeisler (2026b), together with patch-micro balanced accuracy and class-specific patient-macro recalls. Probabilities are evaluated as produced; post-hoc temperature scaling is not applied.

Three contrasts summarize the surface, all expressed in percentage points:

$$\Delta_m(G) = A(G, 32) - A(G, 8), \quad \text{depth gain at fixed breadth,} \quad (5)$$

$$\Delta_G(m) = A(20, m) - A(5, m), \quad \text{breadth gain at fixed depth,} \quad (6)$$

$$X = A(20, 8) - A(5, 32), \quad \text{equal-budget breadth advantage.} \quad (7)$$

The equal-budget advantage X compares the broadest and the deepest allocation of the $n_c = 160$ set and is the most informative single quantity in the design, because both of its terms consume the same labelling budget and differ only in how that budget is spread over patients. The intermediate cell (10, 16) completes the set and indicates whether the exchange between breadth and depth is monotone.

4.1 Support surface

The contrasts describe individual comparisons, whereas the accounts of Table 2 concern the grid as a whole. Cell accuracies are therefore also summarized by least-squares fits over the nine cells, using the candidate predictors of Equation (3) on a logarithmic scale,

$$A = \alpha + \beta \log n, \quad A = \alpha + \beta \log G, \quad A = \alpha + \beta \log N_{\text{eff}}, \quad (8)$$

which have equal numbers of parameters and can be ranked directly by residual standard deviation. Two augmented fits then separate the accounts. Adding breadth to the nominal predictor, $A = \alpha + \beta \log n + \gamma_n \log G$, asks whether patient count matters at all once the patch budget is known, and so reproduces the benchmark finding within this design. Adding breadth to the effective predictor, $A = \alpha + \beta \log N_{\text{eff}} + \gamma_e \log G$, asks the question this experiment exists for: whether patients still matter once their redundancy has been accounted for. The second fit nests the third model of Equation (8) at $\gamma_e = 0$.

Uncertainty follows the procedure of the two preceding experiments. Each of 2,000 replicates of a paired Bayesian bootstrap assigns random positive weights drawn from a $\text{Dirichlet}(1, \dots, 1)$ distribution to the distinct test patients (Rubin, 1981), with every patient’s weight shared across cells, draws, splits, classes, and repeated test appearances. Within-class recalls, cell scores, contrasts, and surface coefficients are recomputed on each replicate, and the 2.5th and 97.5th percentiles form individual exploratory 95% intervals without adjustment for multiple comparisons. These intervals describe held-out patient variation conditional on the realized training draws; variation from the draws themselves is reported separately as the between-draw dispersion of Section 2.1.

4.2 Interpretation criteria

One percentage point remains the prespecified practical-relevance threshold (Queisler, 2026e). A point estimate above one point is practically promising, a lower interval bound above one point supports an effect exceeding the threshold, and an interval crossing zero leaves the direction open. For the surface coefficients the threshold is expressed on the same scale by noting that doubling breadth changes $\log G$ by $\log 2$, so a coefficient matters practically when $\gamma \log 2$ exceeds one point, that is when γ exceeds 1.44 points per log unit.

The accounts are then read as follows. Support that depends only on the patch budget requires both X and γ_n to be indistinguishable from zero, which the benchmark result makes unlikely and which therefore functions as a check on the design rather than as a live hypothesis. Redundancy-adjusted support is supported when the effective predictor attains the smallest residual standard deviation of the three fits in Equation (8) and the interval for γ_e lies inside the practical band around zero. Coverage beyond redundancy is supported when the interval for γ_e excludes zero with a positive point estimate, and is practically relevant when that estimate also exceeds 1.44 points per log unit. Because both augmented fits are conditional on the estimated correlations, a positive γ_e can also indicate that the scalar projection underestimates within-patient dependence, so the class-level correlation table is consulted before the residual term is read as missing patient variation.

Depth saturation is judged separately from the fits. It is supported when $\Delta_m(20)$ has an interval contained in ± 1 point, that is when quadrupling the patches taken from each of twenty patients leaves accuracy practically unchanged. This criterion is informative only if the predicted transition falls inside the ladder, so the mean estimated correlation is compared with the depth levels in advance: a mean ICC below $1/32$ places the transition beyond the deepest cell, and an absence of saturation is then uninformative about the redundancy account rather than evidence against it.

5 Results

Broader patient sampling improves accuracy at a fixed patch budget in both datasets. On BRACS, additional patches from the same patients also improve accuracy, and effective support gives the closest single-predictor description of the grid. On TCGA-UT, depth gains are small, breadth dominates, and a positive residual breadth term remains after adjustment for effective support.

5.1 Realized allocations and accuracy grid

Both common-pool feasibility audits passed without reducing the grid. Eligible patients per class and split range from 26 to 56 on BRACS and from 30 to 514 on TCGA-UT. These counts are identical across all three depth levels. All 135 selected fits per dataset converged, covering nine cells, three splits, and five draws. Their pooled patient-macro balanced accuracies reproduce the completed analyses. Table 3 reports the full grid, with each cell averaging fifteen fitted allocations.

BRACS			
G	$m = 8$	$m = 16$	$m = 32$
5	34.55 [32.24, 36.91]	35.73 [33.16, 38.27]	37.13 [34.50, 39.78]
10	37.72 [35.25, 40.22]	38.69 [35.67, 41.70]	39.21 [36.08, 42.27]
20	40.84 [38.21, 43.71]	41.68 [39.16, 44.39]	42.88 [40.10, 45.71]

TCGA-UT			
G	$m = 8$	$m = 16$	$m = 32$
5	61.97 [60.73, 63.09]	62.13 [60.90, 63.23]	62.25 [61.02, 63.34]
10	67.04 [65.78, 68.25]	67.28 [66.01, 68.51]	67.39 [66.10, 68.58]
20	71.76 [70.43, 72.94]	72.02 [70.68, 73.20]	72.08 [70.75, 73.26]

Table 3: Patient-macro balanced accuracy (%) and paired-bootstrap 95% intervals. G denotes patients per class and m patches per patient. Scores average splits and draws equally; intervals condition on the realized training allocations.

BRACS accuracy rises from 34.55% at (5, 8) to 42.88% at (20, 32); TCGA-UT rises from 61.97% to 72.08%. Point estimates increase with both factors throughout both grids. At 160 patches per class, BRACS accuracy progresses from 37.13% at (5, 32) through 38.69% at (10, 16) to 40.84% at (20, 8). The corresponding TCGA-UT sequence is 62.25%, 67.28%, and 71.76%. Thus the intermediate allocation also favours breadth at this budget; the conclusion does not depend only on the two extreme cells.

5.2 Equal-budget advantage and depth gains

The equal-budget breadth advantage is 3.71 points on BRACS (95% interval: 2.21 to 5.35) and 9.52 points on TCGA-UT (8.96 to 10.07). Both lower bounds exceed the one-point practical threshold. These comparisons contradict a description based only on nominal patch count within the tested allocation procedure. They compare patient composition and depth at a fixed budget, with validation-selected regularization in each cell.

BRACS		TCGA-UT	
Contrast	Estimate [95% interval]	Contrast	Estimate [95% interval]
$\Delta_m(5)$	2.58 [1.87, 3.38]	$\Delta_m(5)$	0.28 [0.16, 0.41]
$\Delta_m(10)$	1.49 [0.46, 2.43]	$\Delta_m(10)$	0.34 [0.23, 0.45]
$\Delta_m(20)$	2.04 [1.13, 2.93]	$\Delta_m(20)$	0.32 [0.24, 0.41]
$\Delta_G(8)$	6.30 [4.78, 7.98]	$\Delta_G(8)$	9.80 [9.24, 10.36]
$\Delta_G(16)$	5.95 [4.39, 7.78]	$\Delta_G(16)$	9.88 [9.31, 10.45]
$\Delta_G(32)$	5.75 [4.34, 7.16]	$\Delta_G(32)$	9.84 [9.27, 10.41]
X	3.71 [2.21, 5.35]	X	9.52 [8.96, 10.07]

Table 4: Accuracy contrasts in percentage points, with paired 95% intervals. $\Delta_m(G)$ quadruples depth from 8 to 32 at fixed breadth; $\Delta_G(m)$ quadruples breadth from 5 to 20 at fixed depth. X exchanges depth for breadth at 160 patches per class.

BRACS depth gains remain positive at each breadth (Table 4). At twenty patients per class, quadrupling depth improves accuracy by 2.04 points (1.13 to 2.93), establishing a gain above one point and failing the prespecified saturation criterion. On TCGA-UT, the same contrast is 0.32 points (0.24 to 0.41). This is a supported positive effect, but its entire interval lies within the practical equivalence band of ± 1 point. All three TCGA-UT depth-gain intervals fall inside this band. Practical saturation here means little additional accuracy over the tested depth range, not exactly zero benefit.

Breadth gains at fixed depth range from 5.75 to 6.30 points on BRACS and from 9.80 to 9.88 points on TCGA-UT, with every lower interval bound above one point. These gains also increase the patch budget fourfold. The equal-budget contrast is therefore the stronger evidence that allocation over patients matters beyond the number of labelled patches.

5.3 Effective support and residual breadth

The mean projected intraclass correlation is 0.130 on BRACS and 0.457 on TCGA-UT. Mean design effects consequently rise from 1.91 to 5.02 over depths 8–32 on BRACS, and from 4.20 to 15.17 on TCGA-UT. At $G = 20$, class-averaged effective support rises from 86.94 to 140.26 on BRACS, but only from 38.92 to 43.58 on TCGA-UT. The correlation model therefore predicts a much weaker return from additional depth in TCGA-UT, consistent with its small observed depth gains.

BRACS		
Predictor	Slope [95% interval]	Residual SD
$\log n$	2.90 [2.30, 3.54]	1.32
$\log G$	4.33 [3.32, 5.45]	0.98
$\log N_{\text{eff}}$	4.31 [3.36, 5.33]	0.30
TCGA-UT		
Predictor	Slope [95% interval]	Residual SD
$\log n$	3.66 [3.46, 3.87]	3.12
$\log G$	7.10 [6.69, 7.50]	0.18
$\log N_{\text{eff}}$	7.07 [6.66, 7.47]	0.25

Table 5: Single-predictor support surfaces over nine pooled cell scores. Slopes are percentage points per natural log unit; residual standard deviations are percentage points, using seven residual degrees of freedom. Ranking describes in-sample fit, not predictive validation.

On BRACS, effective support gives the smallest residual standard deviation among the three single-predictor models: 0.30 points, compared with 0.98 for breadth alone and 1.32 for nominal support (Table 5). Adding breadth to nominal support leaves $\gamma_n = 2.86$ points per log unit (1.74 to 4.07). Adding it to effective support instead gives $\gamma_e = 0.13$ (−1.60 to 1.85). This residual is compatible with zero, but its interval extends outside the practical band of approximately ± 1.44 points per log unit. The BRACS grid is therefore well described by effective support without establishing that its residual breadth effect is practically negligible.

On TCGA-UT, breadth alone fits slightly more closely than effective support (residual SD 0.18 versus 0.25 points); nominal support fits poorly at 3.12 points. The residual after effective adjustment is $\gamma_e = 4.30$ points per log unit (3.56 to 4.99), equivalent to 2.98 points (2.47 to 3.46) for a doubling of breadth at fixed effective support in the fitted model. Its lower bound exceeds the practical threshold. This supports a residual breadth association beyond this particular scalar adjustment. The grid contains no exact matched-effective-support intervention, so that conditional coefficient remains a model-based summary rather than a direct matched comparison.

Dataset	Adjustment	γ [95% interval]	Residual SD
BRACS	Nominal	2.86 [1.74, 4.07]	0.29
BRACS	Effective	0.13 [-1.60, 1.85]	0.32
TCGA-UT	Nominal	6.87 [6.46, 7.28]	0.12
TCGA-UT	Effective	4.30 [3.56, 4.99]	0.12

Table 6: Residual breadth coefficients from the two-predictor models, with paired 95% intervals. Coefficients are points per natural log unit; residual SD uses six residual degrees of freedom. The slope of the support predictor is jointly refitted in each model.

The reciprocal mean correlations place the approximate transition depths near 7.7 on BRACS and 2.2 on TCGA-UT. Both are at or below the smallest depth, but the BRACS gain shows that this transition is not a sharp accuracy plateau. Effective support still increases by about 61% from depth 8 to 32 on BRACS, compared with 12% on TCGA-UT. A design-effect transition alone does not imply that accuracy has saturated.

5.4 Split consistency, draw variation, and reporting scope

Equal-budget breadth gains are positive in every split, ranging from 3.49 to 3.99 points on BRACS and from 8.73 to 10.37 on TCGA-UT (Table 7). Depth gains at twenty patients range from 1.70 to 2.60 points on BRACS and from 0.27 to 0.37 on TCGA-UT. These split-level results preserve the pooled contrast between useful additional depth on BRACS and small depth gains on TCGA-UT.

Dataset / split	X	$\Delta_m(20)$	$\Delta_G(8)$	Draw SD range
BRACS / 0	3.65	1.70	6.23	0.33–3.11
BRACS / 1	3.99	2.60	6.75	0.79–2.60
BRACS / 2	3.49	1.82	5.90	1.35–3.58
TCGA-UT / 0	9.46	0.32	9.81	0.34–1.94
TCGA-UT / 1	8.73	0.37	9.09	0.48–1.03
TCGA-UT / 2	10.37	0.27	10.49	0.30–1.48

Table 7: Split-level contrasts in points, averaging five draws per cell. Draw SD is calculated separately within each split and cell using divisor five; the displayed range spans its nine cells. This separates training-draw dispersion from differences between splits. Splits overlap and are not independent cohort replications.

Draw variation is appreciable, particularly on BRACS. Within-split, within-cell standard deviations span 0.33–3.58 points on BRACS and 0.30–1.94 on TCGA-UT. These are descriptive dispersions over five training draws, not intervals for a fresh training cohort; the paired-bootstrap intervals above describe held-out patient variation conditional on those draws.

Class correlations are reported in Table 8. BRACS ranges from 0.077 for usual ductal hyperplasia to 0.252 for invasive carcinoma; TCGA-UT ranges from 0.314 for lung squamous cell carcinoma to 0.641 for kidney renal clear cell carcinoma. These estimates characterize dependence along the leading principal direction, not all morphological heterogeneity. The completed analysis summarizes patient-macro balanced accuracy but does not contain the prespecified macro negative log-likelihood, expected calibration error, patch-micro balanced accuracy, or class-specific patient-macro recalls. Those secondary endpoints remain unreported here; no conclusion about probability quality, calibration, or class-specific recovery follows from the accuracy results.

6 Discussion

At the tested patch budget, allocating material across more patients is beneficial in both cohorts. Equal contributions remove unequal loss weight as an explanation, while common eligibility pools and nested depth samples prevent depth from changing the selected patients. The remaining contrast concerns how much additional evidence comes from more material from the same patients and how much comes from a broader patient sample. BRACS benefits from both; TCGA-UT gains substantially more from breadth, with less than half a point from quadrupling depth at any tested breadth.

The BRACS findings make redundancy-adjusted effective support a useful descriptive summary. Its fit is substantially closer than nominal support or breadth alone, and adding a residual breadth term gives little improvement. Nevertheless, the uncertainty interval does not establish practical equivalence to zero. It would therefore be premature to conclude that redundancy fully explains the established deficit, or that deeper sampling can generally replace additional patients. Even within this experiment, the direct equal-budget comparison favours breadth by more than two points at its lower interval bound.

TCGA-UT combines high projected within-patient dependence, small depth gains, and a positive residual breadth coefficient after effective-support adjustment. This combination shows that the chosen scalar adjustment does not fully summarize the accuracy surface. It is compatible with additional patients supplying conditions absent from the observed ones, but also with dependence missed by the projection or misspecification of the log-linear surface. The result does not establish which morphological or acquisition factors matter, nor that no amount of additional depth could ever compensate for patient shortage.

For dataset construction, the supported implication is restricted but concrete: within these common eligible pools, twenty patients contributing eight patches each outperform five contributing 32 each. The contrast holds across all fixed splits and both cohorts. Extending that recommendation to other tissue-availability distributions, representations, or budgets requires further allocation experiments. Multiple feature projections and repeated cohorts would help determine whether the residual breadth term reflects limitations of the redundancy measure or patient variation that requires a richer account.

7 Limitations

The common eligibility rule restricts training to patients with at least $m_{\max}$ patches of the relevant class. These patients may differ systematically from patients with less tissue or fewer slides. Pairing patients and nesting patches removes changes in eligibility and sampled patient identity from within-draw depth contrasts, but does not establish generalization to excluded patients. Validation and test populations remain unchanged. Depth effects also depend on the deterministic slide-balanced patch ordering used here.

The grid is bounded by the rarest class in each cohort, so its largest allocation stays well below the benchmark’s balanced allocations and the absolute accuracies of this experiment are not comparable with those of the earlier reports. The design also fixes equal depth for every patient, which removes contribution inequality but makes the allocations less similar to naturally collected cohorts, where patients differ widely in how much material they supply. Conclusions about the shape of the support surface therefore transfer to naturally unequal cohorts only through the design-effect model.

The redundancy adjustment inherits the approximations of the taxonomy. Its correlation is estimated from a one-dimensional projection of the representation under an exchangeable within-patient correlation model, so dependence that the leading principal direction does not express is absorbed into the residual breadth term. The support surface itself is a descriptive least-squares summary over nine cells; it establishes which predictor orders the observations, not a causal or generative account of the learning problem. All estimates are conditional on frozen Virchow2 features, on multinomial logistic regression, and on the three fixed splits, whose partitions overlap across splits.

Finally, the experiment locates the deficit in the training material but cannot say whether the missing information is absent from the cohort or merely inexpressible in this representation. A residual breadth term is compatible with both, and separating them requires varying the representation rather than the allocation. The prespecified probability-quality endpoints are not summarized in the completed analysis. Discrimination gains therefore cannot be interpreted as improvements in calibration.

A Class-level correlations

The class estimates below average the three training-split correlations. Design effects follow Equation (3). They describe dependence along the leading principal direction and do not quantify all between-patient variation in the representation.

BRACS			
Class	ICC _c	DE _c (8)	DE _c (32)
Normal	0.109	1.77	4.39
Pathological benign	0.168	2.18	6.22
Usual ductal hyperplasia	0.077	1.54	3.39
Flat epithelial atypia	0.100	1.70	4.11
Atypical ductal hyperplasia	0.084	1.58	3.59
Ductal carcinoma in situ	0.118	1.82	4.65
Invasive carcinoma	0.252	2.76	8.80
TCGA-UT			
Class	ICC _c	DE _c (8)	DE _c (32)
Brain Lower Grade Glioma	0.429	4.00	14.29
Breast invasive carcinoma	0.448	4.14	14.88
Glioblastoma multiforme	0.337	3.36	11.44
Lung adenocarcinoma	0.398	3.78	13.32
Lung squamous cell carcinoma	0.314	3.20	10.74
Uterine Corpus Endometrial Carcinoma	0.485	4.40	16.04
Sarcoma	0.493	4.45	16.27
Head and Neck squamous cell carcinoma	0.329	3.30	11.20
Kidney renal clear cell carcinoma	0.641	5.48	20.86
Thyroid carcinoma	0.520	4.64	17.12
Bladder Urothelial Carcinoma	0.508	4.56	16.76
Stomach adenocarcinoma	0.452	4.16	15.00
Skin Cutaneous Melanoma	0.462	4.23	15.31
Prostate adenocarcinoma	0.527	4.69	17.34
Liver hepatocellular carcinoma	0.521	4.64	17.14
Colon adenocarcinoma	0.394	3.76	13.21
Kidney renal papillary cell carcinoma	0.564	4.95	18.49
Cervical squamous cell carcinoma and endocervical adenocarcinoma	0.515	4.61	16.97
Testicular Germ Cell Tumors	0.493	4.45	16.29
Adrenocortical carcinoma	0.500	4.50	16.51
Pancreatic adenocarcinoma	0.328	3.29	11.16
Thymoma	0.358	3.50	12.08
Esophageal carcinoma	0.410	3.87	13.71
Kidney Chromophobe	0.489	4.42	16.16
Ovarian serous cystadenocarcinoma	0.513	4.59	16.91
Uterine Carcinosarcoma	0.378	3.64	12.71
Mesothelioma	0.414	3.90	13.85
Rectum adenocarcinoma	0.530	4.71	17.42
Uveal Melanoma	0.329	3.30	11.19
Pheochromocytoma and Paraganglioma	0.636	5.45	20.72

Table 8: Class-specific intraclass correlations and implied design effects at depths 8 and 32. Larger values indicate stronger projected within-patient dependence.

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